Time Series - Forecasting

Advanced Statistics and Alternative Data Types

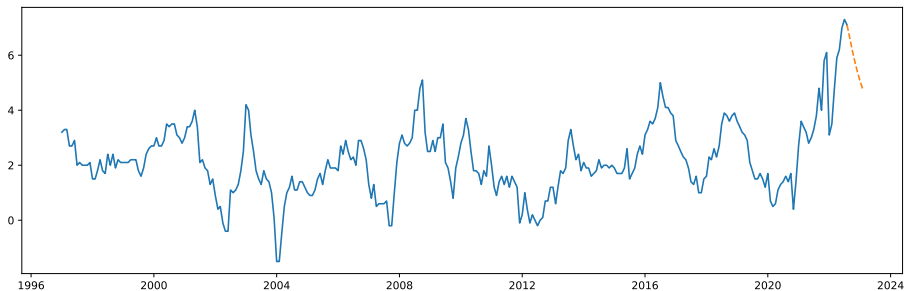
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Forecasting

- One of the central goals of time series analysis is to forecast
  - We have observations on a time series $X_1, \ldots, X_T$ up to period T
  - We want to forecast or predict the value X_{T+h} at $T + h$ for $h \geq 1$

- Below we plot the time series of inflation in Norway and its forecasted values for the next half year using an AR(2) model Code



Optimal forecasts

- We saw earlier in the course that the optimal prediction in terms of MSE is the conditional expectation:
$$\hat{X}_{T+h} = \mathbb{E}[X_{T+h}|X_1, \ldots, X_T].$$
- This did not rely on any sampling assumptions and so also applies in time series
- However, working with the conditional expectation is in general difficult
- We typically instead work with the best linear prediction
  - The exception is for Gaussian time series
  - In the Gaussian case, the conditional expectation coincides with the best linear prediction

Best linear predictor

- The best linear prediction of a random variable X with $\mathbb{E}X^2 < \infty$ in terms of random variables $W = (W_1, \ldots, W_k)$ with each $\mathbb{E}W_k^2 < \infty$ is the linear combination of $W_1, \ldots, W_K$ which minimises the mean squared error:

$$MSE = \mathbb{E}\left[X - \sum_{k=1}^{K} \lambda_k W_k\right]^2$$

- We will write the best linear predictor as $E[X|W]$.
  - Note the difference between $E[X|W]$ and the conditional expectation $\mathbb{E}[X|W]$.

Best linear predictor

Proposition

Let $X, W_1, \ldots, W_K$ be random variables such that each $\mathbb{E}W_k^2 < \infty$ and $\mathbb{E}X^2 < \infty$. Define

$$\hat{X} = \sum_{k=1}^{K} \lambda_k W_k, \qquad \text{where} \qquad \sum_{k=1}^{K} \lambda_k \mathbb{E}[W_k W_j] = \mathbb{E}[X W_j] \qquad \text{for } j = 1, \ldots, K.$$

Then $\hat{X} = E[X|W]$, i.e. $\hat{X}$ is the best linear predictor of X given W:

$$\hat{X} = \operatorname{argmin}_{Z \in \operatorname{span}(W_1, \ldots, W_K)} \mathbb{E}[X - Z]^2.$$

Best linear predictor

Proof.

Let $W = (W_1, \ldots, W_K)$ and $\lambda = (\lambda_1, \ldots, \lambda_K)$ and note that for any $\alpha \in \mathbb{R}^K$, we have

$$\begin{aligned}
\mathbb{E}\left[X - \alpha' W\right]^2 &= \mathbb{E}\left[X - \lambda' W + \lambda' W - \alpha' W\right]^2 \\
&= \mathbb{E}\left[X - \lambda' W\right]^2 + 2\mathbb{E}\left[(X - \lambda' W) W'\right](\lambda - \alpha) + \mathbb{E}\left[\lambda' W - \alpha' W\right]^2 \\
&= \mathbb{E}\left[X - \lambda' W\right]^2 + \mathbb{E}\left[\lambda' W - \alpha' W\right]^2,
\end{aligned}$$

which is minimised when $\alpha = \lambda$. □

Best linear predictor

Remark

$\lambda = (\lambda_1, \ldots, \lambda_K)$ is defined by the equations:

$$\sum_{k=1}^{K} \lambda_k \mathbb{E}[W_k W_j] = \mathbb{E}[X W_j] \qquad \text{for } j = 1, \ldots, K$$

Re-writing these in matrix form shows that if $\mathbb{E}[WW']$ is nonsingular then

$$\lambda = \mathbb{E}[WW']^{-1}\mathbb{E}[WX].$$

Properties of best linear predictor

Proposition

Let X, Y be random variables such that $\mathbb{E}X^2 < \infty$, $\mathbb{E}Y^2 < \infty$, $W = (W_1, \ldots, W_K)$ such that $\mathbb{E}\|W\|^2 < \infty$. Also let $\alpha \in \mathbb{R}^K$. Then

1. $E[X|W] = \lambda' W$ where λ satisfies $\mathbb{E}[(X - \lambda' W)W] = 0$,
2. $E[\alpha_1 X + \alpha_2 Y|W] = \alpha_1 E[X|W] + \alpha_2 E[Y|W]$,
3. $E[\alpha' W|W] = \alpha' W$,
4. $E[X|W] = E[E[X|W, V], W]$ if $\mathbb{E}\|V\|^2 < \infty$.

- We will prove (2) and (4). (1) and (3) are left as exercises.

Properties of best linear predictor

Proof.

2. Let λ_1 and λ_2 be such that $\mathbb{E}[(X - \lambda_1' W)W] = 0$ and $\mathbb{E}[(Y - \lambda_2' W)W] = 0$. Then also $\mathbb{E}[(\alpha_1 X - \alpha_1 \lambda_1' W)W] = 0$ and $\mathbb{E}[(\alpha_2 Y - \alpha_2 \lambda_2' W)W] = 0$ and so

$$\mathbb{E}[(\alpha_1 X - \alpha_1 \lambda_1' W + \alpha_2 Y - \alpha_2 \lambda_2' W)W] = 0.$$

4. Suppose V is a random vector of dimension M and let $\lambda_1 \in \mathbb{R}^K$ and $\lambda_2 \in \mathbb{R}^M$ be such that $E[X|W, V] = \lambda_1' W + \lambda_2' V$. By (2) $E[\lambda_1' W + \lambda_2 V|W] = \lambda_1' W + \beta' W$ where $\beta' W = E[\lambda_2' V|W]$. Since β satisfies $\mathbb{E}[(\lambda_2' V - \beta' W)W] = 0$,

$$\begin{aligned}
\mathbb{E}\left[(X - (\lambda_1 + \beta)' W)W\right] &= \mathbb{E}\left[(X - \lambda_1' W)W\right] - \mathbb{E}\left[(\beta' W)W\right] \\
&= \mathbb{E}\left[(X - \lambda_1' W - \lambda_2' V)W\right] \\
&= 0.
\end{aligned}$$

□

Best linear predictor: AR(2)

Example

Suppose that $(X_t)_{z\in\mathbb{Z}}$ is a causal AR(2):

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t, \quad \epsilon_t \sim \mathrm{WN}(0, \sigma^2).$$

We can use [2] & [3] of the Proposition to calculate that the best linear predictor of X_{T+1} given $X_1, \ldots, X_T$ is

$$\begin{aligned} E[X_{T+1}|X_1, \ldots, X_T] &= E[\phi_1 X_T + \phi_2 X_{T-1} + \epsilon_{T+1}|X_1, \ldots, X_T] \\ &= \phi_1 X_T + \phi_2 X_{T-1} + E[\epsilon_{T+1}|X_1, \ldots, X_T] \\ &= \phi_1 X_T + \phi_2 X_{T-1}, \end{aligned}$$

as $E[\epsilon_{T+1}|X_1, \ldots, X_T] = 0$ because $\mathbb{E}[\epsilon_{T+1} X_l] = 0$ for each $l \in \{1, \ldots, T\}$.[a]

[a] To see this use the causal representation of X_t as a MA(∞) process.

Calculating best linear predictors

- The preceding example can be extended analogously to any AR(p) [see the notes]
- Calculating best linear predictors for MA or ARMA models is more complex
- Fortunately there are algorithms which allow recursive calculation of best linear predictors:

$$\hat{X}_1 = 0,\ \hat{X}_2 = E[X_2|X_1],\ \hat{X}_3 = E[X_3|X_1, X_2],\ \ldots,\ \hat{X}_{T+1} = E[X_{T+1}|X_1, \ldots, X_T].$$

- We will discuss two:
  - The Durbin – Levinson algorithm
  - The innovations algorithm

Durbin – Levinson algorithm

Proposition

Suppose that $(X_t)_{t\in\mathbb{Z}}$ is a zero-mean stationary process with autocovariance function γ. Suppose that $\Gamma_n = \mathrm{Var}(X^n)$ is positive definite for each $n \in \mathbb{N}$, where $X^n = (X_1, \ldots, X_n)'$,

$$E(X_{n+1}|X^n) = \lambda_{n,1}X_n + \cdots + \lambda_{n,n}X_1, \quad \text{and} \quad \nu_n = \mathbb{E}\left[X_{n+1} - E(X_{n+1}|X^n)\right]^2.$$

$\lambda_{n,j}$ and ν_n can be computed as follows: $\lambda_{1,1} = \gamma(1)/\gamma(0)$, $\nu_0 = \gamma(0)$,

$$\lambda_{n,n} = \left[\gamma(n) - \sum_{j=1}^{n-1} \lambda_{n-1,j}\gamma(n-j)\right] \nu_{n-1}^{-1}, \qquad \nu_n = \nu_{n-1}[1 - \lambda_{n,n}^2],$$

$$\begin{bmatrix} \lambda_{n,1} \\ \vdots \\ \lambda_{n,n-1} \end{bmatrix} = \begin{bmatrix} \lambda_{n-1,1} \\ \vdots \\ \lambda_{n-1,n-1} \end{bmatrix} - \lambda_{n,n} \begin{bmatrix} \lambda_{n-1,n-1} \\ \vdots \\ \lambda_{n-1,1} \end{bmatrix}.$$

Partial autocorrelations

- The function $\alpha : \mathbb{N} \to \mathbb{R}$ defined by $\alpha(n) = \lambda_{n,n}$ is the partial autocorrelation function.
- This is the correlation between X_{n+1} and X_1 after controlling linearly for the intermediate observations $X_2, \ldots, X_n$.

Proposition

The partial autocorrelation function (PACF) α satisfies

$$\alpha(k) = \lambda_{k,k} = \mathrm{Corr}(X_{k+1} - E[X_{k+1}|(X_2, \ldots, X_k)], X_1 - E[X_1|(X_2, \ldots, X_k)]).$$

Innovations algorithm

- The innovations algorithm is more generally applicable than the Durbin – Levinson algorithm — it applies also to non-stationary processes
- Define $\hat{X}_1 = 0$ and then let $\hat{X}_{n+1} = E[X_{n+1}|X^n]$. Since each $\hat{X}_l$ is a linear combination of $X_1, \ldots, X_{l-1}$, we have $\text{span}(X_1, \ldots, X_n) = \text{span}(X_1 - \hat{X}_1, \ldots, X_n - \hat{X}_n)$, so that there are $\theta_{n,j}$ for $n \in \mathbb{N}$ and $j \in \{1, \ldots, n\}$ such that

$$\hat{X}_{n+1} = \sum_{j=1}^{n} \theta_{n,j}(X_{n+1-j} - \hat{X}_{n+1-j}) \tag{1}$$

- Recall that

$$\nu_n = \mathbb{E}\left[X_{n+1} - E(X_{n+1}|X^n)\right]^2 \tag{2}$$

Innovations algorithm

Proposition

Suppose that $(X_t)_{t\in\mathbb{Z}}$ is a zero-mean process with autocovariance function γ and $\Gamma_n = \mathrm{Var}(X^n)$ is positive definite for each $n \in \mathbb{N}$. Then $\theta_{n,j}$ and ν_n can be found from

$$\theta_{n,n-k} = \nu_k^{-1}\left(\Gamma(n+1,k+1) - \sum_{j=0}^{k-1}\theta_{k,k-j}\theta_{n,n-j}v_j\right) \quad 0 \le k < n,$$

and

$$\nu_n = \Gamma(n+1,n+1) - \sum_{j=0}^{n-1}\theta_{n,n-j}^2\nu_j \quad n \ge 1, \qquad \nu_0 = \Gamma(1,1).$$

h-step ahead forecasts using the innovations algorithm

- The innovations algorithm directly gives us a 1-step ahead forecast:[1]

$$\hat{X}_{T+1} = E[X_{T+1}|X^T] = \sum_{j=1}^{T} \theta_{T,j}(X_{T+1-j} - \hat{X}_{T+1-j})$$

- It can also be used to construct h-step ahead forecasts. For $h \geq 1$,

$$\begin{aligned} E[X_{T+h}|X^T] &= E\left[E\left[[X_{T+h}|X^{T+h-1}\right]|X^T\right] \\ &= E[\hat{X}_{T+h}|X^T] \\ &= E\left[\sum_{j=1}^{T+h-1} \theta_{T+h-1,j}(X_{T+h-j} - \hat{X}_{T+h-j})\middle| X^T\right] \\ &= \sum_{j=h}^{T+h-1} \theta_{T+h-1,j}(X_{T+h-j} - \hat{X}_{T+h-j}). \end{aligned}$$

[1] As does the Durbin – Levinson algorithm.

Best linear predictor: MA(1)

Example

Suppose that $(X_t)_{t\in\mathbb{Z}}$ is an MA(1) process. We will calculate the coefficients of the best linear prediction of X_t given $X_{t-1},\dots,X_1$ using the innovations algorithm. We have

$$X_t = \epsilon_t + \theta\epsilon_{t-1}, \quad \epsilon_t \sim \mathrm{WN}(0,\sigma^2).$$

By a result earlier in the course, $\Gamma(i,j) = 0$ for $|i-j|>1$, $\Gamma(i,i) = \sigma^2(1+\theta^2)$ and $\Gamma(i,i+1) = \sigma^2\theta$. Applying the innovations algorithm we obtain $\nu_0 = \sigma^2(1+\theta^2)$, then $\theta_{1,1} = \sigma^2\theta/\sigma^2(1+\theta^2)$ and $\nu_1 = \sigma^2(1+\theta^2) - \theta^2\sigma^2/(1+\theta^2)$. Letting $n \geq 2$, we have that for all $2 \leq j \leq n$, $\theta_{n,j} = 0$, and

$$\theta_{n,1} = \nu_{n-1}^{-1}\theta\sigma^2, \qquad \nu_n = \sigma^2(1+\theta^2) - v_{n-1}^{-1}\theta^2\sigma^4.$$

Innovations algorithm for ARMA processes

- The innovations algorithm can be simplified if $(X_t)_{t\in\mathbb{Z}}$ is an ARMA(p,q) process.
- This is achieved by instead applying the algorithm to the process

$$W_t = \begin{cases} \sigma^{-1}X_t & t = 1,\dots,m \\ \sigma^{-1}\Phi(L)X_t & t > m \end{cases}, \qquad m = \max(p,q).$$

Proposition

Suppose that $(X_t)_{t\in\mathbb{Z}}$ is an ARMA(p,q) process. Suppose that $\Gamma_n = \mathrm{Var}(X^n)$ is positive definite for each $n \in \mathbb{N}$. Then,

$$\hat{X}_{n+1} = \begin{cases} \sum_{j=1}^{n} \theta_{n,j}(X_{n+1-j} - \hat{X}_{n+1-j}) & \text{for } 1 \le n < m \\ \sum_{r=1}^{p} \phi_r X_{n+1-r} + \sum_{j=1}^{q} \theta_{n,j}(X_{n+1-j} - \hat{X}_{n+1-j}) & \text{for } n \ge m \end{cases},$$

and $\mathbb{E}(X_{n+1} - \hat{X}_{n+1})^2 = \sigma^2\mathbb{E}(W_{n+1} - \hat{W}_{n+1})^2 = \sigma^2 r_n$, where $r_n = \mathbb{E}(W_{n+1} - \hat{W}_{n+1})^2$ are calculated during the application of the innovations algorithm to the W_t.

h-step ahead forecasts for ARMA processes

Remark

Suppose that $(X_t)_{t\in\mathbb{Z}}$ is an ARMA(p,q) process. By properties of the best linear predictor

$$E[X_{t+h}|X^t] = \begin{cases} E\left[X_{n+h}|X^n\right], & h \le m-n \\ E\left[\Phi(L)X_{n+h}|X^n\right], & h > m-n \end{cases}$$

$$= \begin{cases} \sum_{j=h}^{n+h-1} \theta_{n+h-1,j}(X_{n+h-j} - \hat{X}_{n+h-j}), & h \le m-n \\ \sum_{r=1}^{p} \phi_r E[X_{n+h-r}|X^n] + \sum_{j=h}^{n+h-1} \theta_{n+h-1,j}(X_{n+h-j} - \hat{X}_{n+h-j}), & h > m-n \end{cases}.$$

If $n > m = \max(p,q)$ then for all $h \ge 1$, $\theta_{n+h-1,j} = 0$ for $j > q$ and hence

$$E[X_{t+h}|X^t] = \sum_{r=1}^{p} \phi_r E[X_{n+h-r}|X^n] + \sum_{j=h}^{q} \theta_{n+h-1,j}(X_{n+h-j} - \hat{X}_{n+h-j}).$$

Looking ahead

- In pratice most software packages include functions which can forecast h-steps ahead after a model has been fitted
- Nevertheless it's important to understand what the Durbin – Levinson & innovations algorithms do
- We will see next week that:
  1. Estimating ARMA models often relies on the innovations algorithm for reasons of computational efficiency
  2. How the ACF and PACF can guide us about which ARMA model to choose

Code for AR2 forecast

```
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm


df = pd.read_csv(...)
df.index = pd.PeriodIndex(df.index.str.strip().str.replace("M", "-") + "-01", freq = "M").to_timestamp()
X = pd.to_numeric(df["CP00"], errors= "coerce")[0:-1]
Xhat = sm.tsa.AutoReg(X, 2, trend="c").fit().forecast(steps = 6)
Xhat = pd.to_numeric(pd.concat((pd.DataFrame(X[-2:]).iloc[1:], pd.DataFrame(Xhat, columns=["CP00"])))["CP00"])
fig, ax = plt.subplots(figsize = (12, 4))
ax.plot(X, color = "tab:blue")
ax.plot(Xhat, color = "tab:orange", linestyle = "--")
plt.savefig("output/inflation_forecast.pdf")
```

◂ Forecasting